

LLaVA-CoT: first Vision Language Model with Step-by-Step Reasoning capabilities similar to GPT-o1



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Current Vision-Language Models (VLMs) often struggle to perform systematic and structured reasoning, especially when handling complex visual question-answering tasks. To address it, this paper introduces LLaVA-CoT, a novel VLM designed to conduct autonomous multistage reasoning. Unlike chain-of-thought prompting, LLaVA-CoT independently engages in sequential stages of summarization, visual interpretation, logical reasoning, and conclusion generation.

Key contributions:

- introduce LLaVA-CoT, a visual language model designed for systematic reasoning, demonstrating exceptional performance on tasks that require structured thinking and reasoning.
- demonstrate that LLaVA-CoT, using stage-level beam search, is inference-time scalable. This means that with increased computational resources, the performance of our approach can be further enhanced, making it applicable to more complex tasks and scenarios.
- experiments on various benchmarks demonstrate that our method achieves superior performance relative to larger and closed-source models, underscoring the effectiveness of LLaVA-CoT for multimodal reasoning

Proposed Method

LLaVA-CoT facilitates a progressive, step-by-step reasoning process that enhances the reasoning capabilities of Vision-Language Models (VLMs) and allows for effective inference time scaling.

i) Enhancing Reasoning Capability through Structured Thinking

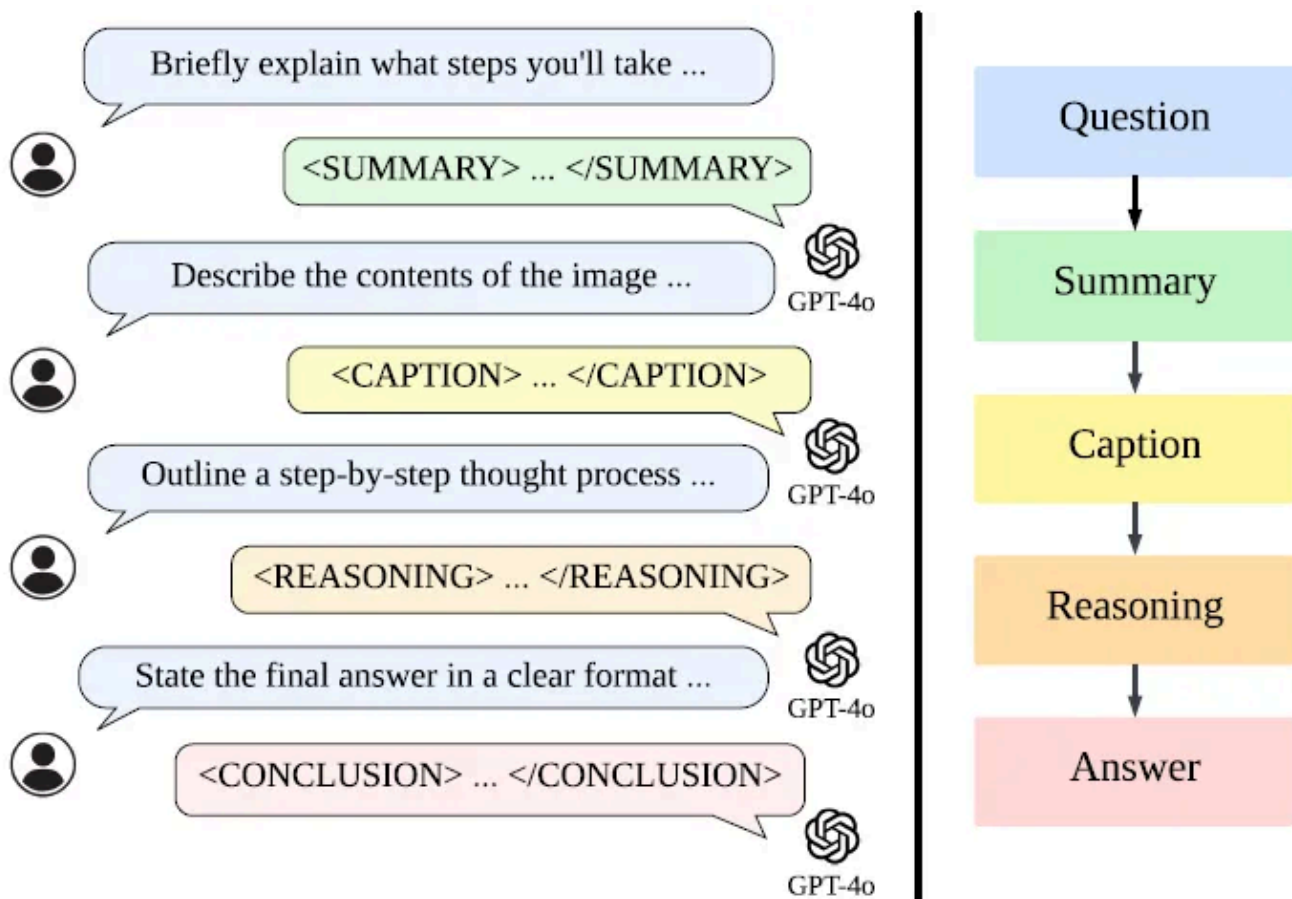
a) Reasoning Stages

LLaVA-CoT, decomposes the answer generation process into four structured reasoning stages:

- **Summary Stage:** In this initial phase, LLaVA-CoT provides a high-level summary interpretation of the question, outlining the primary aspects of the problem it intends to address.
- **Caption Stage:** If an image is present, LLaVA-CoT offers a concise overview of the visual elements relevant to the question, helping to understand multimodal input.
- **Reasoning Stage:** Building on the initial summary, LLaVA-CoT conducts structured, logical reasoning to derive a preliminary answer.
- **Conclusion Stage:** In this final stage, LLaVA-CoT synthesizes an answer based on the preceding reasoning. Here, the output from the conclusion stage is the direct response provided to the user, while the prior three stages are internal "hidden stages" representing LLaVA-CoT's reasoning process.

b) Data Preparation

- compiled a new dataset, integrating samples from several widely used VQA datasets, resulting in a total of 99k image QA pairs
- Figure below shows Process flow for generating the LLaVA-CoT-100k dataset, where authors prompted GPT-4o to generate responses in separate stages, and filter its outputs to ensure quality.



- also included data from both general-purpose VQA datasets and science-targeted VQA datasets specified below:

(1) General VQA Datasets

- ShareGPT4V, which provides multi-turn question-answering data from GPT-4V interactions
- ChartQA, which focuses on interpreting charts and graphs
- A-OKVQA, which emphasizes external knowledge beyond visible content.
- DocVQA, which involves document-based questions requiring textual comprehension.
- PISC, which helps to understand social relationships, and CLEVR [24] to address object properties, spatial relationships, and counting tasks.

(2) Science-Targeted VQA Datasets

- GeoQA+ for geometric reasoning, along with AI2D and ScienceQA, which target scientific questions

(3) CLEVR-Math

- an extension of CLEVR, focuses on arithmetic analysis in visual contexts.

Table below shows the number of QA pairs selected from each dataset

Dataset	Type	Size
ShareGPT4V [9]	General VQA	31.3k
ChartQA [40]	General VQA	17.2k
A-OKVQA [48]	General VQA	16.1k
AI2D [25]	Science-Targeted VQA	11.4k
GeoQA+ [7]	Science-Targeted VQA	11.4k
ScienceQA [36]	Science-Targeted VQA	5.6k
DocVQA [41]	General VQA	4.0k
PISC [30]	General VQA	1.0k
CLEVR [24]	General VQA	0.5k
CLEVR-Math [14]	Science-Targeted VQA	0.5k

c) Model Training

- LLaVA-CoT-100k dataset constructed can be used to further conduct Supervised Fine-Tuning (SFT) on any existing model to enhance reasoning capabilities
- In this work, authors select the Llama-3.2-11B Vision-Instruct model as the base model, and perform a full parameter fine-tuning by using the LLaVA-CoT-100k dataset.

ii) Effective Inference Time Scaling using Stage-level Beam Search

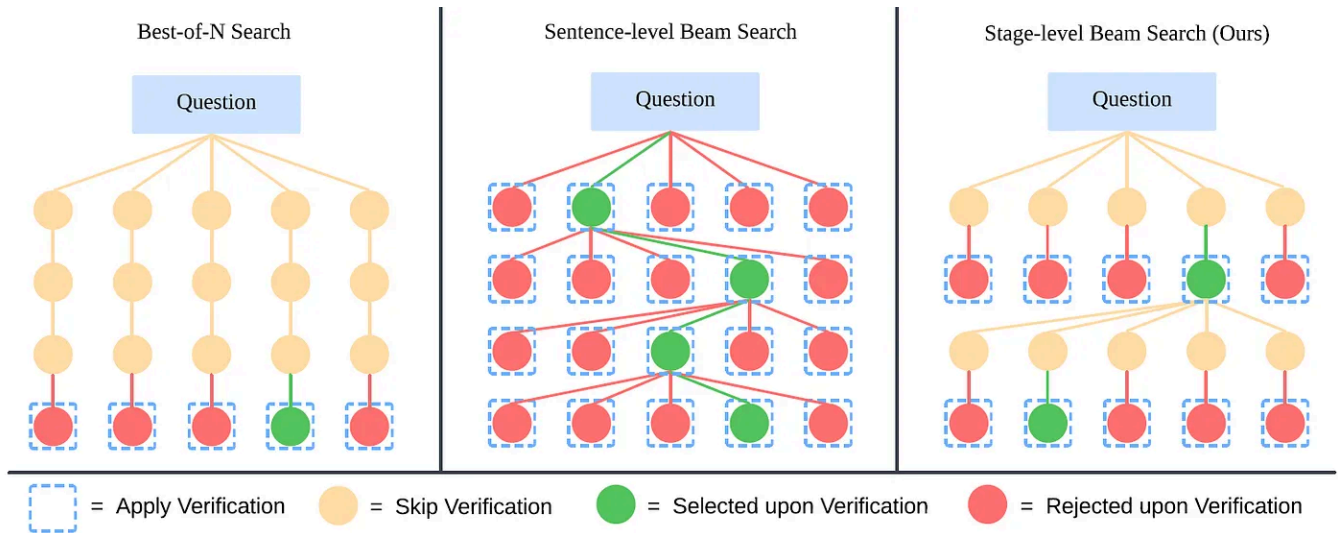
leverage the stage-based outputs of LLaVA-CoT, which provides an ideal granularity for inference time scaling. Our method follows the steps below:

- Sample N responses for the first stage in the solution.
- Randomly sample 2 responses and let the model determine which is better, keeping the better response.
- Repeat for N – 1 times, retaining the best response.

- Sample N responses for the next stage, then repeat steps 2–4 until all stages are processed.

It is the structured output design of LLaVA-CoT that makes this approach feasible, enabling efficient and accurate verification at each stage.

Illustration of the three approaches is shown in Figure below:



- Best-of-N search generates N complete responses and selects the best one among them
- Sentence-level Beam Search generates multiple candidate options for each sentence and chooses the best one. In contrast, our Stagelevel
- Beam Search generates candidates for each reasoning stage (e.g., summary, caption, reasoning, and conclusion) and selects the best option at each stage.
- Best-of-N search operates at a coarse level, while Sentence-level Beam Search is overly granular, and our method achieves an optimal balance and achieves the best performance.

Figure below shows Comparison of LLaVA-CoT performance with and without stage-level beam search. Our stage-level beam search is effective in selecting better reasoning during model inference.



Question: What is the total price for a bottle of Merlot and a bottle of Cabernet shown in the image?

<SUMMARY> To solve the problem, I will first identify the prices of the Merlot and Cabernet bottles shown in the image. Then, I will add these prices together to find the total cost. </SUMMARY>

<CAPTION> The image displays three wine bottles with their respective prices. The first bottle is a Northstar Merlot from Columbia Valley, priced at \$139.99. The second bottle is a Moët & Chandon Imperial from France, priced at \$114.99. The third bottle is a Chateau St. Michelle Cold Creek Cabernet from Columbia Valley, priced at \$109.99. </CAPTION>

Answer without inference scaling:

<REASONING> To find the total price for a bottle of Merlot and a bottle of Cabernet, I will identify the prices of these two specific wines. The Merlot is priced at \$139.99, and the Cabernet is priced at \$109.99. **Adding these two prices together will give the total cost for one bottle of each.** </REASONING>
<CONCLUSION> 149.99 </CONCLUSION>

Answer from beam search:

<REASONING> To find the total price for a bottle of Merlot and a bottle of Cabernet, I will add the price of the Northstar Merlot (\$139.99) to the price of the Chateau St. Michelle Cold Creek Cabernet (\$109.99). **The calculation is as follows: \$139.99 + \$109.99 = \$249.98.** </REASONING>
<CONCLUSION> The total price for a bottle of Merlot and a bottle of Cabernet shown in the image is \$249.98. </CONCLUSION>

Post-Training Performance

i) Benchmark Results

- Table below shows Experimental results of different models on the benchmark. Here, LLaVA-CoT (with Direct Training) refers to the model trained directly on the original VQA dataset's Q&A pairs, while LLaVA-CoT (w/o Structured Tags) represents the model trained on the LLaVA-CoT-100k dataset with the structured tags removed. LLaVA-CoT refers to the model trained on the complete LLaVA-CoT-100k dataset, including the structured tags.

Model	MMStar	MMBench	MMVet	MathVista	AI2D	Hallusion	Average
Base Model							
Llama-3.2-11B-Vision-Instruct	49.8	65.8	57.6	48.6	77.3	40.3	56.6
Our Models							
LLaVA-CoT (with Direct Training)	54.3	76.2	49.9	49.5	91.4	42.9	60.7
LLaVA-CoT (w/o Structured Tags)	55.7	74.2	57.0	54.1	87.2	45.0	62.2
LLaVA-CoT	57.6	75.0	60.3	54.8	85.7	47.8	63.5

- Results show that LLaVA-CoT demonstrates notable improvements across general VQA, mathematical reasoning, scientific VQA, and hallucination control tasks, with an average benchmark score increase of 6.9%, thereby validating the effectiveness of our approach.

Inference Time Scaling

i) Benchmark Results

- Table below shows Experimental results during inference time. LLaVA-CoT (BS = 2) denotes the model using stage-level beam search with a beam size of 2. The results show that stage-level beam search can achieve further significant performance improvements

Model	MMStar	MMBench	MMVet	MathVista	AI2D	Hallusion	Average
Base Model							
Llama-3.2-11B-Vision-Instruct	49.8	65.8	57.6	48.6	77.3	40.3	56.6
Our Models							
LLaVA-CoT	57.6	75.0	60.3	54.8	85.7	47.8	63.5
LLaVA-CoT (BS = 2)	58.1	75.6	61.7	56.1	87.5	48.2	64.5

- findings suggest that stage-level beam search, which is made possible by the structured output design of LLaVA-CoT, is an effective and powerful approach for inference time scaling.

ii) Comparison to Baseline Methods

- Table below shows Comparison to Baseline Methods

Method	Number of Beam	MMVet Score
No Inference Scaling	1	60.3
Best-of-N Search	10	60.9
Sentence-level Beam Search	2	58.4
Stage-level Beam Search	4	62.9

- proposed stage-level beam search outperforms both best-of-N and sentence-level beam search under comparable inference time compute

iii) Scaling Trend of Stage-level Beam Search

- Table below shows Scaling Trend of LLaVA-CoT

Method	Number of Beam	MMVet Score
No Inference Scaling	1	60.3
Stage-level Beam Search	2	61.7
Stage-level Beam Search	3	62.3
Stage-level Beam Search	4	62.9

- As the the number of candidate responses increases, the model's performance consistently improves.

Comparison to State-of-the-Art VLMs

- Table below compare LLaVA-CoT with other state-of-the-art open-source and closed-source vision language models (VLM) across six benchmarks that require advanced reasoning capabilities

Model	MMStar-R	MMBench-R	MMVet-R	MathVista	AI2D	Hallusion	Average
Closed-Source Models							
GPT-4o-0806 [3]	66.0	82.4	80.8	62.7	84.7	54.2	71.8
Claude3.5-Sonnet-0620 [2]	64.2	75.4	68.7	61.6	80.2	49.9	66.7
Gemini-1.5-Pro [46]	56.4	71.5	71.3	57.7	79.1	45.6	63.6
GPT-4o-mini-0718 [43]	54.9	76.9	74.6	52.4	77.8	46.1	63.8
Open-Source Models							
InternVL2-Llama3-76B [11]	68.9	84.4	62.5	65.6	87.6	55.4	70.7
InternVL2-8B [11]	62.5	77.4	56.9	58.3	83.6	45.0	64.0
Ovis1.5-Gemma2-9B [38]	58.7	76.3	50.9	65.6	84.5	48.2	64.0
MiniCPM-V2.6-8B [61]	57.1	75.7	56.3	60.6	82.1	48.1	63.3
Llama-3.2-90B-Vision-Instruct [42]	51.1	76.8	74.1	58.3	69.5	44.1	62.3
VILA-1.5-40B [32]	53.2	75.3	44.4	49.5	77.8	40.9	56.9
Base Model							
Llama-3.2-11B-Vision-Instruct [42]	46.6	64.9	63.8	48.6	77.3	40.3	56.9
Our Models							
LLaVA-CoT	57.5	73.1	66.7	54.8	85.7	47.8	64.3
LLaVA-CoT (BS = 2)	57.8	73.8	71.3	56.1	87.5	48.2	65.8

- results show that LLaVA-CoT consistently outperforms many open-source models of similar or even larger sizes, such as InternVL2-8B, Ovis1.5-Gemma2-9B , MiniCPM-V2.6-8B , Llama-3.2-90B-Vision-Instruct, and VILA-1.5-40B.
- Remarkably, LLaVA-CoT even surpasses certain closed-source models like GPT-4o-mini and Gemini-1.5-pro, underscoring the effectiveness of our structured reasoning approach.

Conclusion

- present LLaVA-CoT, a novel vision language model that performs structured, autonomous reasoning in multiple stages
- contributions are twofold: first, the creation of the LLaVA-CoT-100k dataset with detailed reasoning annotations, which supports training on systematic, structured responses; and second, the proposal of a stage-level beam search method, enabling effective inference time scaling.
- Overall, LLaVA-CoT establishes a new standard for multimodal reasoning in VLMs, offering robust performance and scalability, especially in inference time

Paper: <https://arxiv.org/abs/2411.10440>

Code: <https://github.com/PKU-YuanGroup/LLaVA-CoT>

Model: <https://huggingface.co/Xkev/Llama-3.2V-11B-cot>

Dataset: <https://huggingface.co/datasets/Xkev/LLaVA-CoT-100k>

References:

1. LLaVA-CoT: Let Vision Language Models Reason Step-by-Step by Xu et al.
[arXiv:2411.10440](https://arxiv.org/abs/2411.10440)

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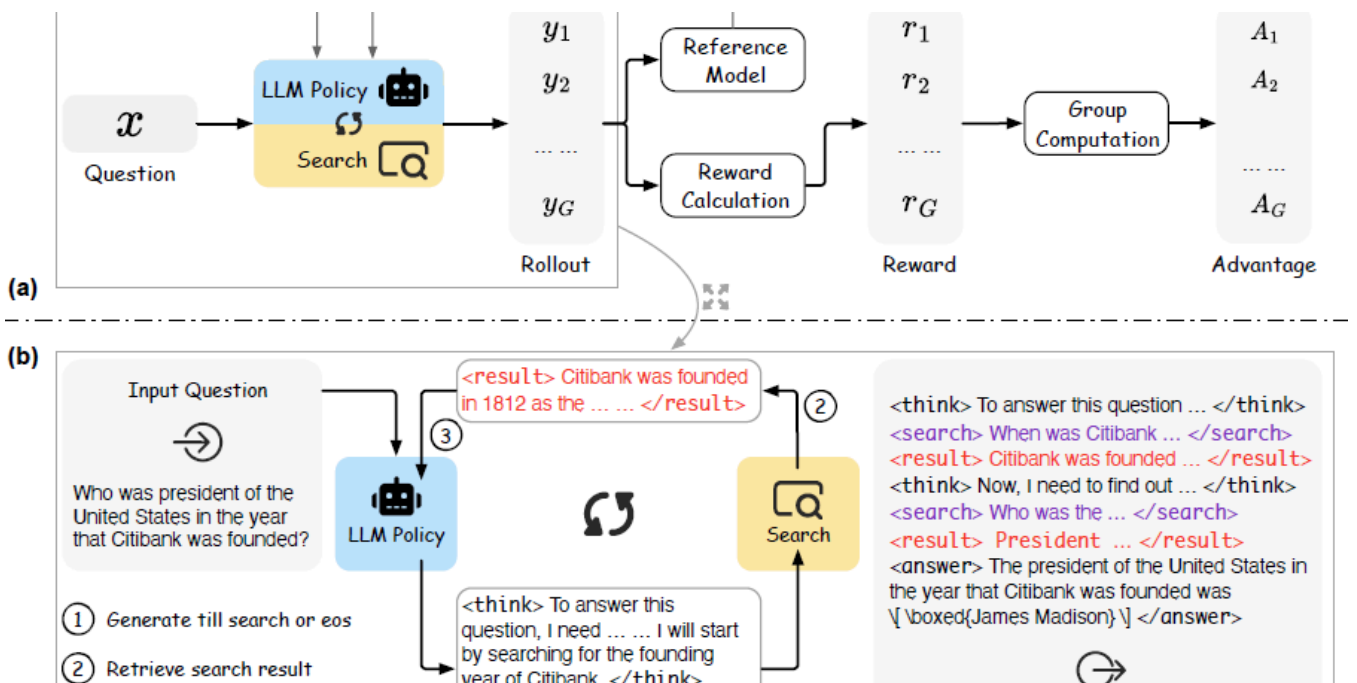
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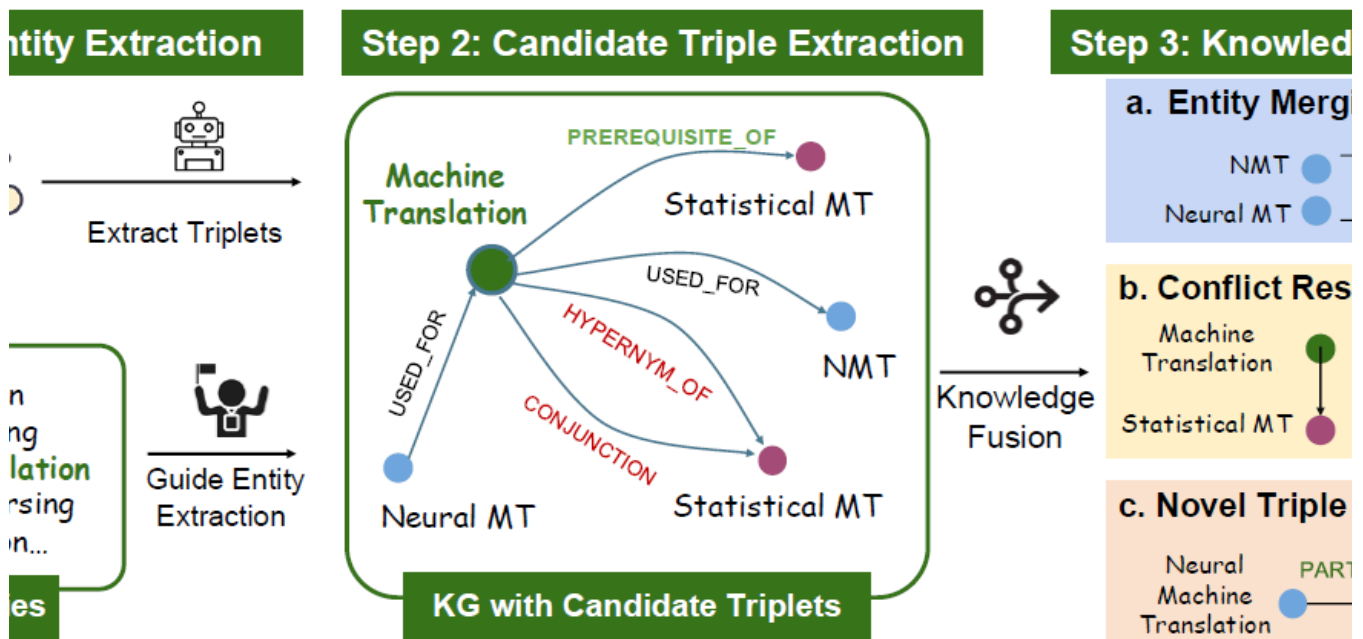


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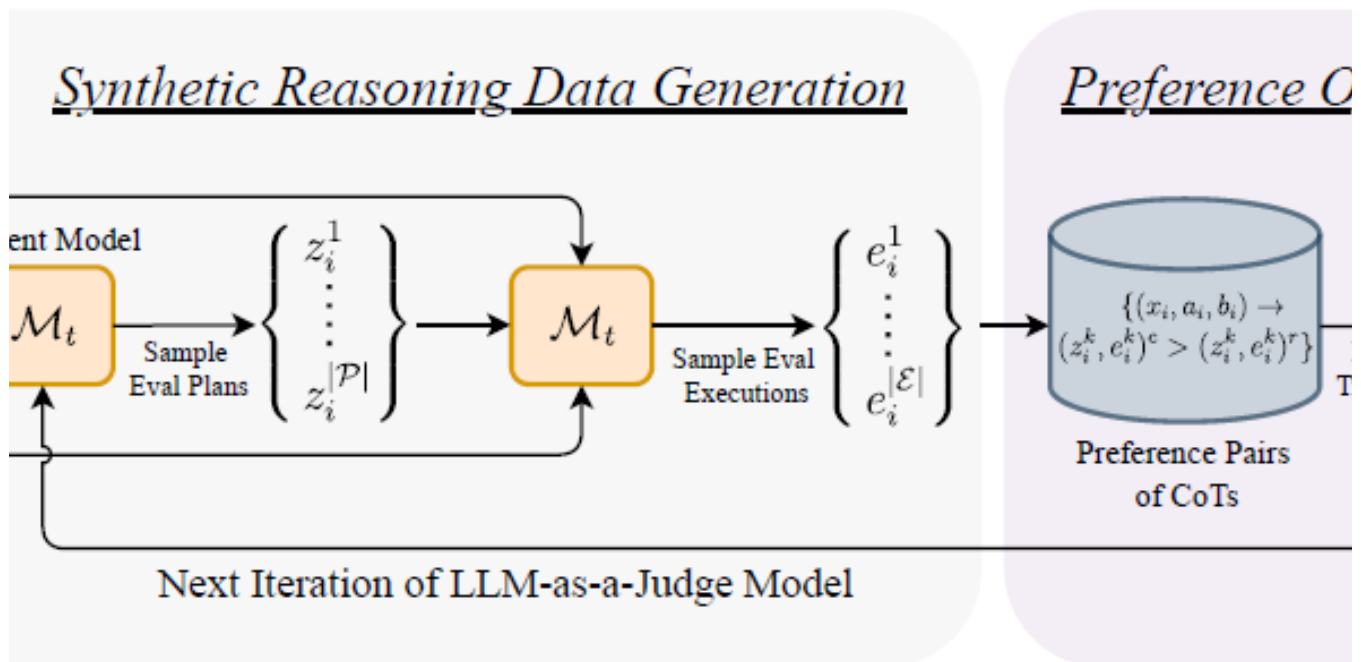


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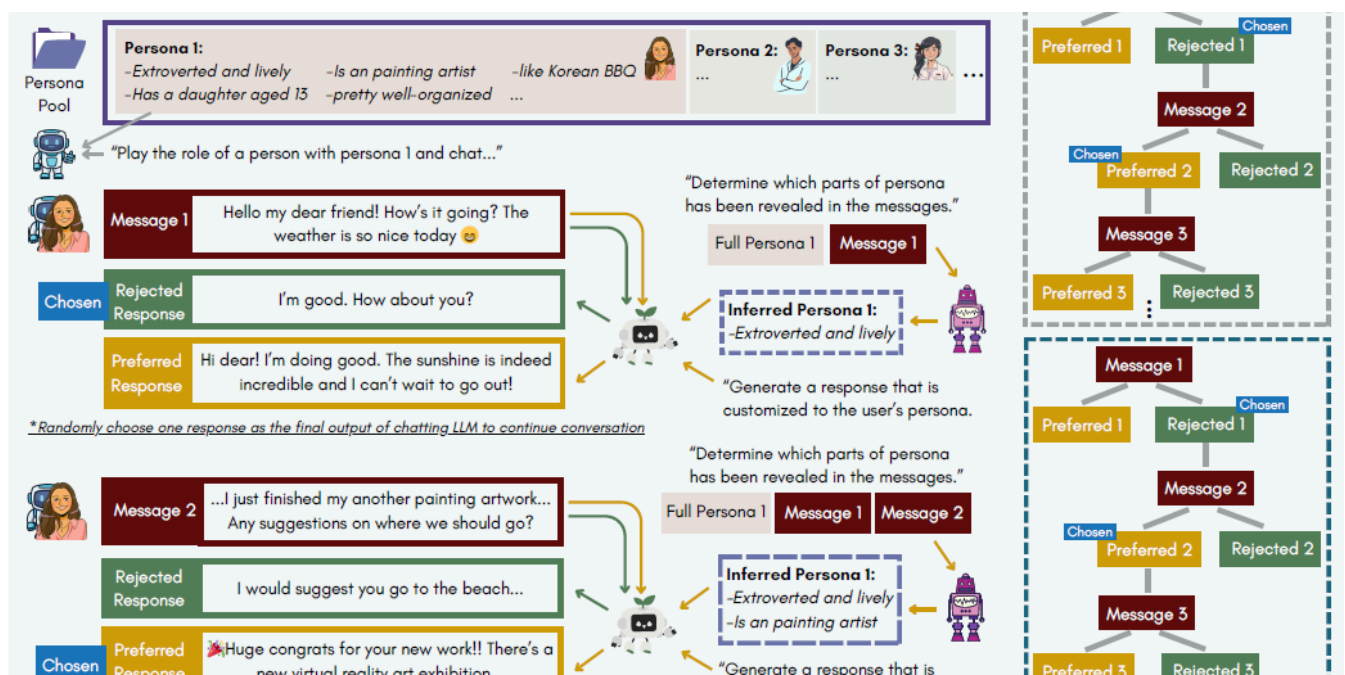


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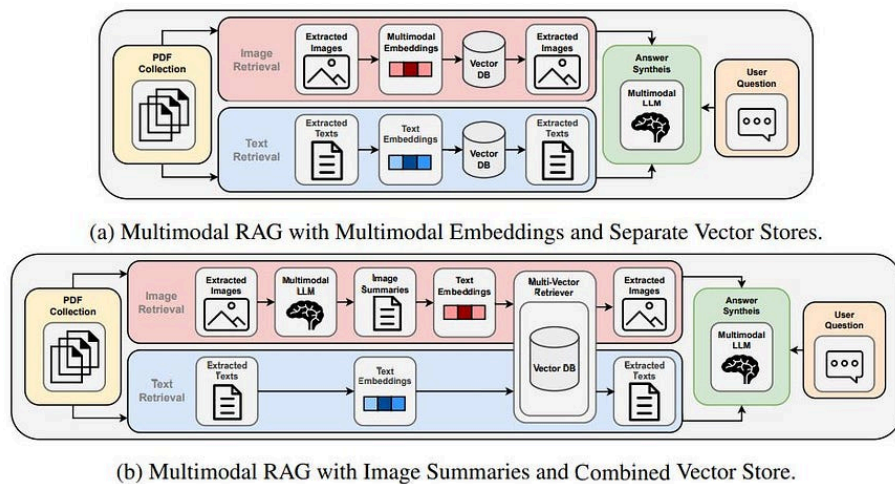
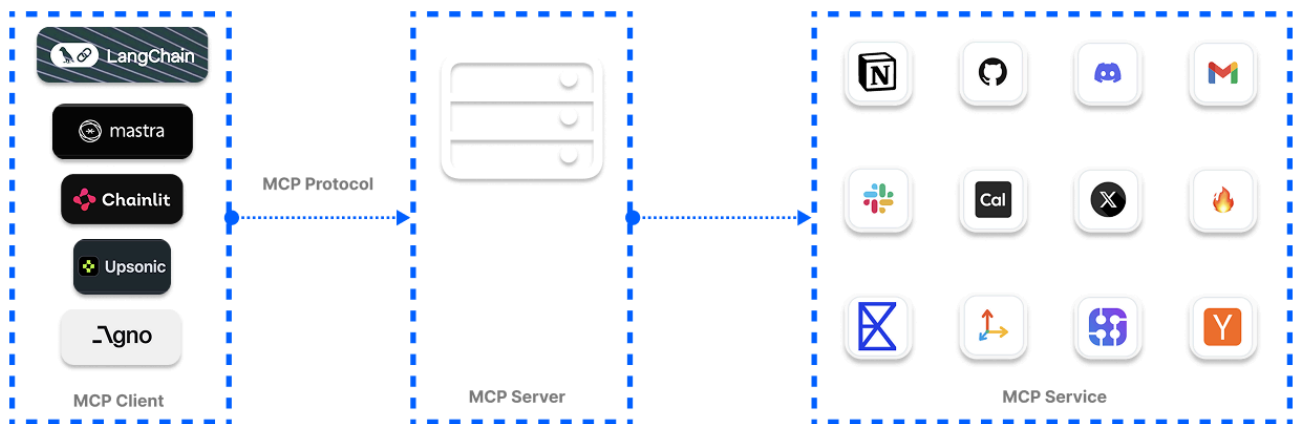


Figure 1: Overall architecture of our proposed multimodal RAG pipelines. For the Text-Only RAG, we only use the **Text Retrieval** component. Conversely, in the Image-Only RAG, we only employ the **Image Retrieval** component, either with multimodal embeddings or image summaries.

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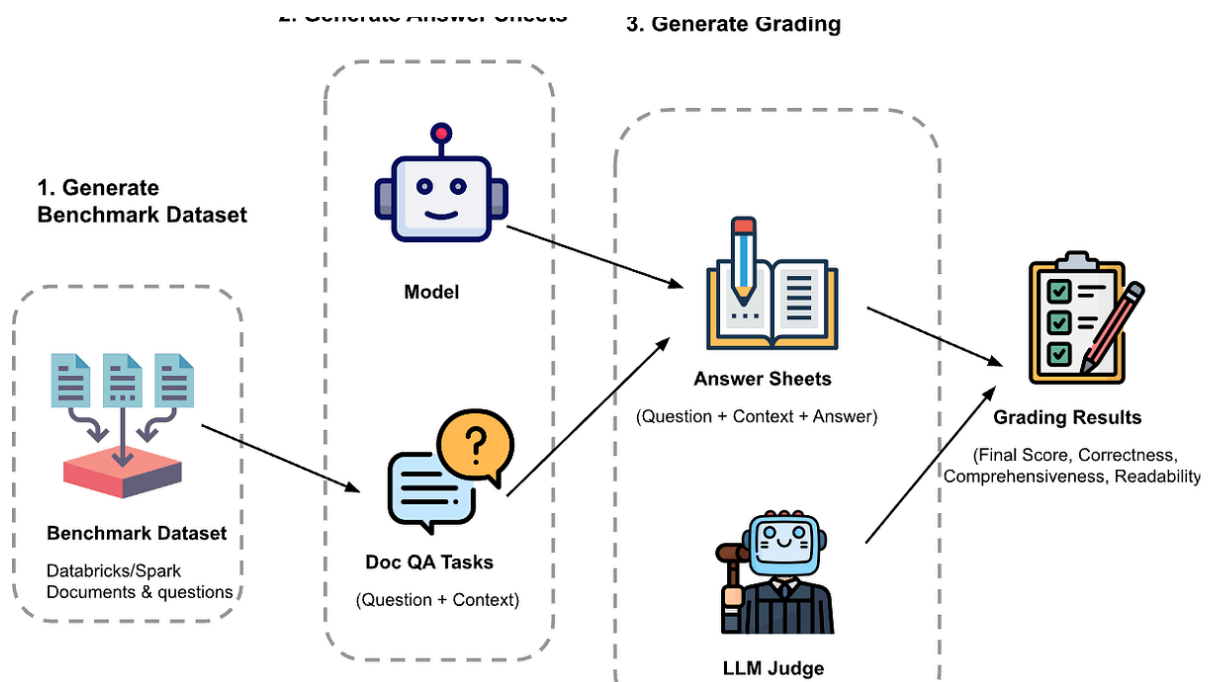


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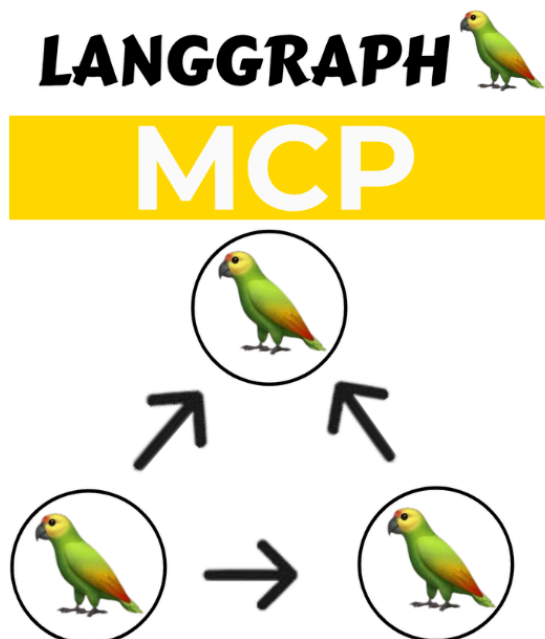
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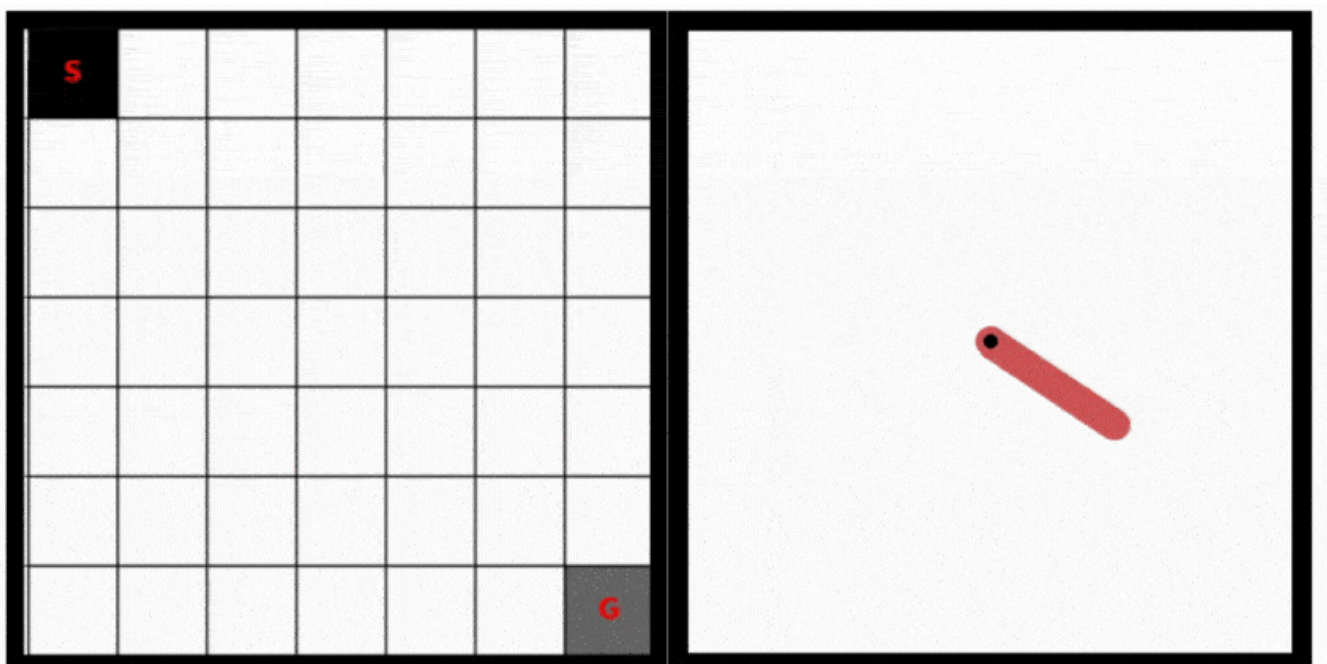
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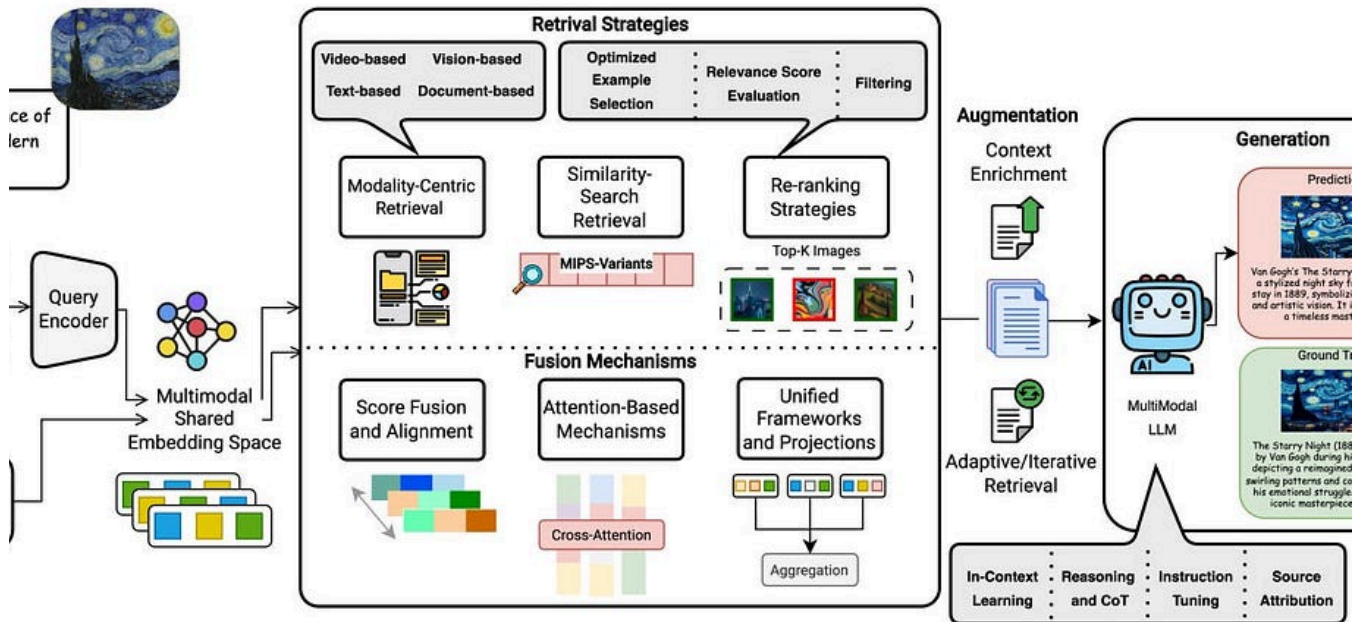
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